LESSON PLAN

NDT (19)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	02
Total No. of Lesson Plans	9	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Liquid penetrant, limitations, physical principles, description of process, penetrant system liquid penetrant materials, types of developer, technique using portable equipment
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Liquid penetrant, limitations, physical principles, description of process, penetrant system liquid penetrant materials, types of developer, technique using portable equipment
5. Specific Objectives	19.1.1.1 Trainee will be able to describe types of NDT and perform liquid penetrant, magnetic particles and eddy current inspections

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of portable liquid penetrant inspection technique; Safety precautions regarding chemical handling and PPE usage; Instructor supervision of trainee application steps
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the three main types of NDT methods?
(b) Why is surface preparation critical before penetrant inspection?
(c) Which NDT method is best for detecting subsurface cracks?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Physical principles of liquid penetrant • Limitations of penetrant inspection • Description of penetrant process steps	L & Ds	PPT, IB	15 min
(b)	• Penetrant system liquid penetrant materials • Types of developer materials • Technique using portable equipment	L & Ds	PPT, PDF	20 min
(c)	• Interpretation of penetrant indications • Cleaning and post-inspection procedures • Quality control checks	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the capillary action principle in penetrant inspection.
(b) List two major limitations of the liquid penetrant method.
(c) Describe the correct sequence for removing excess penetrant.

12. Home Assignment and Follow Up	Prepare a brief report comparing the advantages and limitations of liquid penetrant inspection versus magnetic particle inspection for non-ferrous materials.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Magnetic particle inspection application advantages, limitations, circulars and longitudinal magnetization, general procedure for magnetic particle inspection various types of parts which can be inspected after inspection reasons for demagnetization	Refer to the master syllabus reference list	LP 03

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min